MD SHAHADAT HOSSEN

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SUMMARY

Built and tested autonomous vehicles, with hands-on experience taking a 1/5th-scale autonomous platform from hardware design to outdoor field tests. Now aims to develop heterogeneous aerial, ground, and underwater robots that work as one team, staying coordinated even when communication breaks down, to support search-and-rescue and defense missions.

EDUCATION

M.S. in EE, *GPA: 3.81/4.00* (Aug 2023 – May 2025) Gannon University, Erie, PA, USA
Relevant Coursework: Artificial Neural Networks; Artificial Intelligence; IoT System Fundamentals; Power Electronics; Utility Applications of Power Electronics; Project Management
Achievement: Awarded an $18,000 scholarship and an 18-month graduate assistantship for research excellence.

B.Sc. in EEE, *GPA: 3.86/4.00* (Jan 2017 – Dec 2020) Manarat International University, Dhaka, Bangladesh
Relevant Coursework: Electrical Circuits; Power Systems; Transmission and Distribution; Computer Networks; C/C++ Programming
Achievement: President's Excellence Award for leading innovation in technical workshops and robotics programs.

RESEARCH EXPERIENCE

Graduate Assistant (Autonomous Vehicle Systems & Hardware Testing) Gannon University, Erie, PA, USA
Aug 2023 – May 2025

- Assessed the **cyber-physical system (CPS) framework of a 1/5th-scale autonomous vehicle** and performed **outdoor field tests** to analyze navigation capabilities and system performance.
- Co-authored research on an **integrated sAV/sUAS system for VHF/UHF-band video-based navigation in extreme terrain**, published at **IEEE CCNC 2026**.
- Worked with **ROS, Gazebo, and Ubuntu Linux** to integrate the vehicle's sensors and onboard computer and tested its behavior in simulation before deployment.
- **Diagnosed and resolved electrical and mechanical issues** across motors, wiring, 3D-printed structures, chassis, and compute box, keeping the platform field-ready.
- Collaborated with professors to troubleshoot open problems and **propose new ideas for enhancing the CPS framework**; drafted **documentation guides** mapping its present state and future goals.
- **Mentored** new students through project workflows and helped to resolve technical problems.

Research Assistant Gannon University, Erie, PA, USA
May 2023 – Jun 2023

- Designed and **3D-printed parts for the CPS compute box** using Autodesk Fusion 360 and PrusaSlicer, enhancing its durability and functionality.
- Designed and assembled the **compute box** with strategic selection of electrical components and precise wiring, improving system readiness and operational efficiency.
- Assembled the **chassis with mechanical components and sensors**, resulting in a stable and reliable structure.
- Enhanced **autonomous vehicle navigation accuracy by integrating a GPS module** with satellite systems.

TEACHING EXPERIENCE

Adjunct Lecturer Manarat International University, Dhaka, Bangladesh
May 2022 – Dec 2022

- Taught **three classes of 30 students each**, breaking down complex subject matter into clear, easy lessons.
- Created **lecture slides, video tutorials, and quizzes**, with feedback to help students track their progress and study independently.
- Contributed to **curriculum updates** by integrating emerging technologies and practical applications.

SELECTED PROJECTS

- **Real-Time Face Recognition using a Caffe Model**: Built a real-time face recognition system using OpenCV's deep-learning module with a pre-trained Caffe SSD face detector and an OpenFace embedder generating 128-dimensional face embeddings. Trained an SVM classifier to recognize faces in static images and live webcam video, flagging unknown faces below a confidence threshold. [LINK]
- **Bidirectional Power Flow Control for DC-DC Converters in DC Microgrids**: Designed and simulated a droop-controlled bidirectional DC-DC converter in MATLAB/Simulink to balance power among a solar PV array, battery storage, and load; added MPPT and state-of-charge (SoC)-based droop control, holding about **98%** bus-voltage stability and improving energy harvest by **65%** in simulation. [LINK]

TECHNICAL SKILLS

Robotics & Simulation	ROS, Gazebo, MATLAB/Simulink
Programming Languages	Python, C/C++
ML & Computer Vision	TensorFlow, OpenCV, Scikit-Learn
Design & CAD	Autodesk Fusion 360, AutoCAD, PrusaSlicer, Electronic Workbench
Hardware & Testing	Arduino, Oscilloscope, Digital Multimeter, Power Supply, Soldering
Others	Ubuntu Linux, Visual Studio, PyCharm, XCTU, Microsoft Office

SELECTED PUBLICATIONS

- W. Lee, **M. S. Hossen**, "An Exemplary Platform of Integrated sAV/sUAS System for VHF/UHF-Band Video-Based Navigation in Extreme-Terrain Environments", IEEE Consumer Communications & Networking Conference (CCNC), 2026. [LINK]
- S. Ahmad, S. Forhad, M. H. Shuvo, S. Saifee, **M. S. Hossen**, K. N. I. Nabeen, M. H. Bhuiyan, "Simulated Design of an Autonomous Multi-terrain Modular Agri-bot", International Conference on Machine Learning, IoT and Big Data (ICMIB), Lecture Notes in Networks and Systems, vol. 728, Springer, 2023. [LINK]
- S. Forhad, **M. S. Hossen**, I. A. Ahsan, S. Saifee, K. N. I. Nabeen, M. R. K. Shuvo, "An Intelligent Versatile Robot with Weather Monitoring System for Precision Agriculture", 6th International Conference on Information Systems and Computer Networks (ISCON), IEEE, 2023. [LINK]
- **M. S. Hossen**, T. Ahmed, A. Haque, "Techno-economic Assessment for Installing Small Wind Turbine at Subornochar Region in Bangladesh", 7th Iran Wind Energy Conference (IWEC), IEEE, 2021. [LINK]

HONORS & LEADERSHIP

- **Vice President, IEEE-HKN (Eta Kappa Nu), Iota Nu Chapter** (Aug 2025 – May 2026): Inducted into the honor society for academic excellence in Electrical and Computer Engineering.
- **Treasurer, Bangladesh Student Association, Gannon University** (May 2023 – Nov 2024): Managed event budgets, secured sponsorships, and coordinated event logistics.
- **President, MIU EEE Club** (Jul 2019 – Dec 2020): Organized seminars and industrial tours for student skill development.